MSc Finance 20127

Corporate Finance (Business Valuation)
05 — Leverage, Value, and Optimal Capital Structure

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Where We Are in the Valuation Workflow

Equity story $\longrightarrow$ Financial statements $\longrightarrow$ Forecasts $\longrightarrow$ Cash flows $\longrightarrow$ **Discount rate** $\longrightarrow$
Terminal value $\longrightarrow$ Multiples $\longrightarrow$ Value bridge $\longrightarrow$ Recommendation

Today:
Financing policy $\longrightarrow$ WACC/APV $\longrightarrow$ Enterprise Value $\longrightarrow$ Capital Structure Recommendation

Today's Professional Task

Professional task:

- You advise a firm considering whether to change its leverage
- Your job is to estimate whether the recapitalization creates value
  - estimate how leverage changes cost of equity and cost of debt
  - estimate how leverage changes WACC
  - estimate how leverage changes tax benefits and expected bankruptcy costs
  - identify the debt ratio that maximizes firm value
  - quantify the value consequence of moving to the optimal debt ratio

Plan for Today

Layer	What We Learn	Why It Matters
CORE	How leverage affects value	Capital structure intuition
CORE	WACC approach	Standard corporate finance tool
PRO	Choosing WACC vs APV	Practical valuation/interviews
TECH	Ratings, default spreads, cost of debt	Avoid mechanical WACC analysis
PRO	APV optimization	Leveraged transactions and known debt paths
PRACTICE	Discussion Questions	Turn leverage theory into value impact

From Valuation Method to Financing Decision

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Cost of Capital & APV: Overview

CORE

- Cost of Capital (CoC)
  - Firm value = **discount FCFO at** the Weighted Average Cost of Capital (**WACC**)

    Embedded in firm value are:
    (i) Tax benefits of debt (in the use of the after-tax cost of debt in the WACC)
    (ii) Expected additional risk associated with debt, i.e., higher k_e and k_d at higher D/E ratios

- Adjusted Present Value (APV)
  - Firm value = first **discount FCFO at k_u** (cost of unlevered equity, also called k_a). Then:
    (i) Add PV of interest tax savings generated by debt, and
    (ii) Subtract expected costs of bankruptcy

    Value includes separate estimates of:
    (i) Tax benefits of debt (PV of interest tax savings)
    (ii) Expected additional risk associated with debt (expected bankruptcy cost)

Depending on assumptions on **future growth**, can have 2-stage, 3-stage version
Same as with DDM and FCFE

Cost of Capital & APV: Cost of Capital Approach

CORE

- A firm with FCFO growing at a stable growth rate (steady-state) can be valued as:

$$\text{Firm Value}_0 = \frac{\text{FCFO}_1}{\text{WACC} - g_n}$$

where:

- FCFO_1 = Expected FCFO next year
- WACC = Weighted Average Cost of Capital
- g_n = Annual Growth rate of FCFO forever. Under a stable growth rate assumption:
  - g_n = Reinvestment rate $\times$ Returns on Capital
  - The stable growth rate used in the model has to be $\leq$ the growth rate in the economy (nominal or real depending on CF and WACC)
  - Firm characteristics have to be consistent with assumptions of stable growth

- Limitations
  - Like all stable growth models, it is sensitive to assumptions about g_n
  - Such sensitivity is higher when using WACC (which is lower than k_e)
  - It is sensitive also to assumptions made about CAPEX relative to depreciation
  - Cost of capital should also reflect a stable growth firm (β, D/E)

Cost of Capital & APV: APV Approach

CORE

The APV estimates the **Value of Levered Firm, $V_l = V_u + TB - BC$**, where:

1. **Value of unlevered firm, V_u**

$$V_{u,0} = \frac{\mathbb{E}\left[FCFO_1\right]}{(\rho_u - g_n)} \qquad \text{or} \qquad V_{u,0} = \sum_{t=1}^{n} \frac{\mathbb{E}\left[FCFO_t\right]}{(1+\rho_u)^t} + \frac{\frac{\mathbb{E}[FCFO_{n+1}]}{\rho_{u,st} - g_n}}{(1+\rho_{hg})^n}$$

where:

$\mathbb{E}\left[FCFO_t\right]$: Expected Free Cash Flow from Operations in period t

ρ_u: Unlevered k_e (*hg* high growth; *st* stable)

g_n: Stable growth rate

2. **Expected tax benefit from borrowing, TB**. Two cases:

i. D is low and fixed: $TB_0 = (\tau_C \cdot k_d \cdot D) / k_d = \tau_C \cdot D$

ii. D is high and/or D/E is predetermined: $TB_0 = \sum_{t=1}^{n} \frac{\mathbb{E}[\tau_C \cdot k_d \cdot D]}{(1+\rho_u)^t} + \frac{\tau_C \cdot k_d \cdot D}{(\rho_u - g_n)\cdot(1+\rho_u)^n}$

3. **Expected bankruptcy costs, BC**

$$BC_0 = \pi_d \times LGD$$

where:

π_d: Probability of default

LGD: Loss Given Default

How Leverage Changes Firm Value

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Effect of Leverage on Firm Value: Introduction I

CORE

- Because both the cost of capital and the APV approach make the value of a firm a function of its leverage, it is likely that there is some mix of debt and equity at which firm value is maximized

- The value of the entire firm, using the cost of capital approach, can be estimated by discounting the expected FCFO at the firm's cost of capital:

$$\text{Firm Value}_0 = \sum_{t=1}^{\infty} \frac{\mathbb{E}\left[\text{FCFO}_t\right]}{(1+\text{WACC})^t}$$

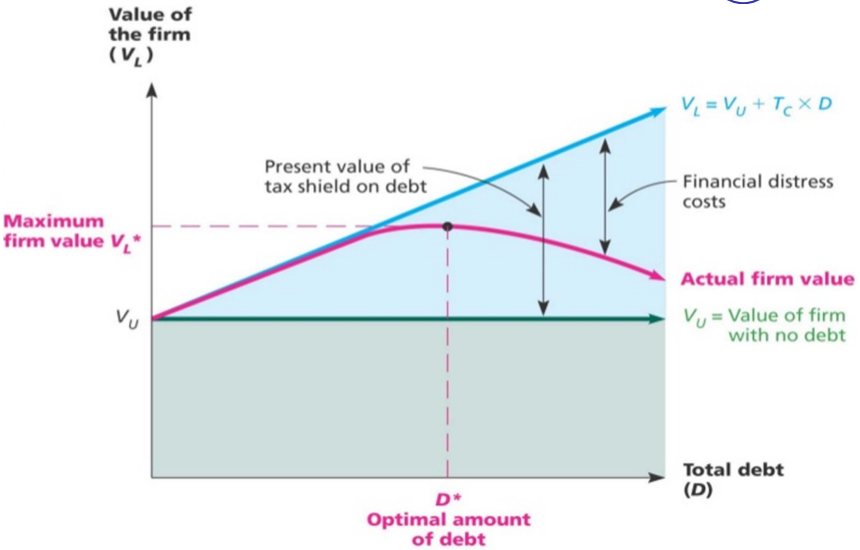
- If we assume that FCFO are unaffected by the choice of financing mix and the cost of capital is reduced as a consequence of increasing debt, the value of the firm will increase
- If the objective in choosing the financing mix for the firm is the maximization of firm value, we can accomplish it by minimizing the cost of capital
- In the more general case where the cash flows to the firm are a function of the debt-equity mix, the optimal financing mix is the mix that maximizes firm value

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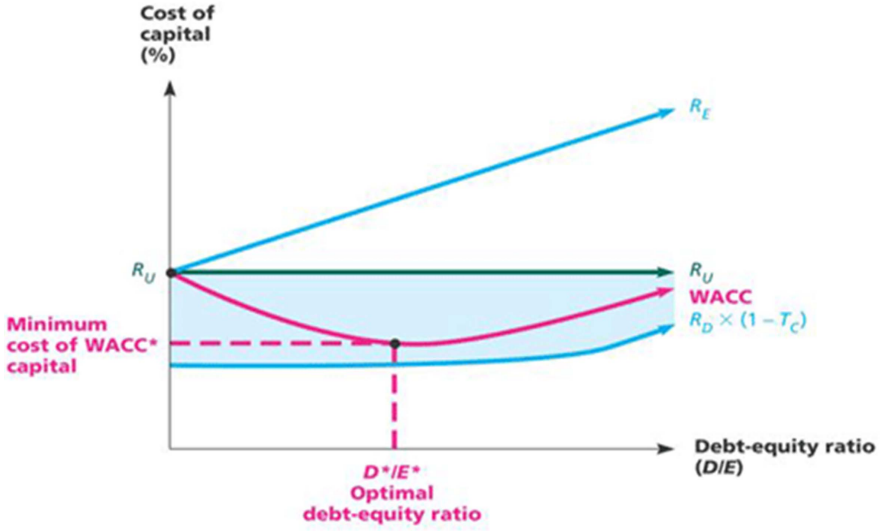
Effect of Leverage on Firm Value: Introduction II

CORE



Effect of Leverage on Cost of Capital: Introduction III

CORE



Finding the Debt Ratio that Maximizes Value

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Optimizing Capital Structure: Cost of Capital Approach I

CORE

Three basic inputs to compute the cost of capital:

1. Cost of Equity
2. (After Tax) Cost of Debt
3. Weights on Debt and Equity

Optimizing Capital Structure: Cost of Capital Approach II

CORE

1. Cost of Equity

- β of equity is a function of (i) the market debt-to-equity ratio and (ii) the unlevered β

$$\beta_{\text{levered}} = \beta_{\text{unlevered}} \times \left[1 + \frac{D}{E}\right]$$

- If we can estimate the unlevered β, we can use it to estimate the levered β at every debt ratio. This levered β can then be used to compute the cost of equity at each debt ratio

$$k_e = \text{Risk Free Rate} + \beta_{\text{levered}} \times \text{Equity Risk Premium}$$

Optimizing Capital Structure: Cost of Capital Approach III

PRO

2. (After Tax) Cost of Debt
   - k_d is a function of the firm's default risk
   - As firms borrow more, default risk will increase and so will k_d

If we use bond ratings to measure default risk, we can estimate k_d in four steps:

a) Estimate (i) a firm's dollar debt and (ii) interest expenses at each debt ratio. Reason: As firms borrow more, both dollar debt and interest expenses will rise

b) At each debt level, (i) compute financial ratio(s) measuring default risk, and (ii) use the ratio(s) to estimate a rating for the firm. Reason: As firms borrow more, rating(s) will decline

c) Add a default spread, based on estimated rating(s), to the risk free rate to get the pre-tax k_d

d) Finally, applying the marginal tax rate yields an after tax k_d

Optimizing Capital Structure: Cost of Capital Approach IV

If Interest Coverage Ratio (Non-financial firms)		Estimated Bond Rating	Default Spread (2026)
Large Cap	Small Cap		
> 8.50	> 12.50	Aaa/AAA	0.69%
[6.50, 8.49]	[9.50, 12.49]	Aa2/AA	0.85%
[5.50, 6.49]	[7.50, 9.49]	A1/A+	1.07%
[4.25, 5.49]	[6.00, 7.49]	A2/A	1.18%
[3.00, 4.24]	[4.50, 5.99]	A3/A-	1.33%
[2.50, 2.99]	[4.00, 4.49]	Baa2/BBB	1.71%
[2.25, 2.49]	[3.50, 3.99]	Ba1/BB+	2.31%
[2.00, 2.24]	[3.00, 3.49]	Ba2/BB	2.77%
[1.75, 1.99]	[2.50, 2.99]	B1/B+	4.05%
[1.50, 1.74]	[2.00, 2.49]	B2/B	4.86%
[1.25, 1.49]	[1.50, 1.99]	B3/B-	5.94%
[0.80, 1.24]	[1.25, 1.49]	Caa/CCC	9.46%
[0.65, 0.79]	[0.80, 1.24]	Ca2/CC	9.97%
[0.20, 0.64]	[0.50, 0.79]	C/C	13.09%
< 0.20	<0.50	D2/D	17.44%

Optimizing Capital Structure: Cost of Capital Approach V

PRO

- Note we only observe promised yield (Yield to Maturity), not the cost of debt k_d
- We can estimate the cost of debt k_d as an internal rate of return (IRR) using expected cash flows (payments) and current price

$$\frac{\mathbb{E}_0\left[\mathrm{CF}_1\right]}{(1+k_d)}+\frac{\mathbb{E}_0\left[\mathrm{CF}_2\right]}{(1+k_d)^2}+\frac{\mathbb{E}_0\left[\mathrm{CF}_3\right]}{(1+k_d)^3}+\ldots+\frac{\mathbb{E}_0\left[\mathrm{CF}_T\right]}{(1+k_d)^T}-\mathrm{Price}_0=0$$

- General Rule : (Promised) Yield to Maturity $\geq$ Cost of Debt
  - Inequality is strict iff there is any probability of default with less than 100% recovery:
    - The difference between the yield to maturity and the cost of debt increases with the probability of default
    - Yield to Maturity $-$ Cost of Debt $=$ Expected Default Loss

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Optimizing Capital Structure: Cost of Capital Approach VI

PRO

3. Weights on Debt and Equity

- The effect on firm value of changing capital structure is isolated by keeping EBIT fixed and varying only the cost of capital. This requires us to make two assumptions:
  - Debt ratio is decreased by raising new equity and retiring debt; conversely, debt ratio is increased by borrowing and buying back stock (recap)
  - Pre-tax EBIT is assumed to be unaffected by the firm's financing mix and, thus, its bond rating
- If EBIT changes with a firm's default risk, the basic analysis won't change, but minimizing WACC may not be optimal, since firm value is determined by both cash flows and WACC
- Firm value will have to be computed at each debt level and the optimal debt ratio will be the one that maximizes firm value

WACC Optimization: The Algorithm

1. Choose candidate debt ratios
2. Relever β at each debt ratio
3. Estimate cost of equity at each debt ratio
4. Estimate interest coverage, rating, spread, and cost of debt at each debt ratio
5. Compute WACC at each debt ratio
6. Revalue the firm at each debt ratio
7. Choose the debt ratio that maximizes firm value

Financial Data for Disney

Inputs	Value
Market Value of Debt	$16,682 million
Market Value of Equity	$45,193 million
Market Price per Share	$24.34
Shares Outstanding	1,856.7 million
Unlevered β	0.7333

Inputs	Value
Treasury Bond Rate	3.5%
Equity Risk Premium	6.0%
Marginal Tax Rate	38.0%
Pre-Tax k_d	6.0%
After Tax k_d	3.72% =6%$\times$(1 $-$ 38%)

	Last Fiscal Year	Trailing 12 Months
Revenues	$37,843	$36,990
EBITDA	$8,986	$8,319
D&A	$1,582	$1,693
EBIT	$7,404	$6,726
Interest Expense	$712	$728
EBITDA (*adjusted for leases*)	$9,989	$8,193
EBIT (*adjusted for leases*)	$7,708	$6,500
Interest Expenses (*adjusted for leases*)	$815	$831
Capital Expenditures (*assumptions*)	$1,628	$1,628

Discussion Questions I

i. What are Disney's current β_{levered}, k_e, and WACC?

ii. What is Disney's β_{levered} at each $D/(D+E)$ ratio?

iii. What is the interest coverage ratio, bond rating, interest rate, and after tax cost of debt, k_d, at each $D/(D+E)$ ratios for Disney?

iv. What is the WACC at each $D/(D+E)$ ratios for Disney?

v. Is Disney currently at its optimal debt ratio? What would be the value consequence of moving to the optimal $D/(D+E)$ ratio for Disney?

APV Optimization: Same Question, Different Decomposition

PRO

WACC Approach	APV Approach
Debt affects WACC	Debt creates tax benefits and distress costs
Find debt ratio minimizing WACC	Find debt ratio maximizing $\mathbf{V}_u + \mathbf{TB} - \mathbf{BC}$
Best when target leverage ratios are stable	Best when dollar debt or debt path is explicit

Optimizing Capital Structure: APV Approach I

CORE

- In the APV approach, we begin with firm value without debt
- Adding debt, we consider benefits and costs of borrowing and their net effect on value
- The value of the levered firm can then be estimated at different levels of debt

The debt level that maximizes firm value is the optimal debt ratio

- The unlevered firm value is independent of expected leverage and can be estimated by discounting FCFO at the unlevered k_U
  - If you do not want to estimate this value and are willing to assume the current market value of the firm is correct, you could back out the unlevered firm value by: (i) subtracting the tax benefits, and (ii) adding back the expected bankruptcy cost from the existing debt

$$\text{Current Firm Value} = \text{Firm Value}_{\text{unlevered}} + PV(TB) - PV(BC)$$

$$\text{Firm Value}_{\text{unlevered}} = \text{Current Firm Value} - PV(TB) + PV(BC)$$

Optimizing Capital Structure: APV Approach II

PRO

- The only components that change as a firm changes its leverage are
  - (i) Expected Tax Benefits
  - (ii) Expected Bankruptcy Costs

  To obtain these values when changing leverage, five steps
  - a) Holding firm value fixed, consider how much dollar debt the firm will have at 20% D/E, 30% D/E, and so on
  - b) Tax benefits of debt = dollar debt $\times$ tax rate; assumes debt and tax benefits are permanent
  - c) Estimate debt rating, interest rate, and interest expense at each debt ratio
  - d) Use the rating to estimate a probability of default (empirical studies)
  - e) Expected bankruptcy cost = probability of bankruptcy $\times$ bankruptcy cost, stated as % of firm value
- You compute the value of the levered firm at different levels of debt

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The debt level that maximizes firm value is the optimal debt ratio

Discussion Questions II

vi. What are the values of tax benefits at each $D/(D+E)$ ratios for Disney?

vii. What are the bond ratings, default probabilities, and expected bankruptcy costs at each $D/(D+E)$ ratios for Disney?

viii. What is the levered firm value at each $D/(D+E)$ ratios for Disney using APV?

Common Mistakes in Capital Structure Optimization

1. Treating more debt as always value-increasing because of tax shields
2. Minimizing WACC while ignoring that operating cash flows may fall with distress risk
3. Using promised yield to maturity as the true cost of debt when default risk is material
4. Holding equity β fixed while changing leverage
5. Holding cost of debt fixed while increasing leverage
6. Using book weights instead of market-value weights without justification
7. Double-counting tax benefits in both WACC and APV
8. Calling the lowest WACC "optimal" without checking the value consequence

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Takeaways

1. Leverage can create value through interest tax shields
2. Leverage can destroy value through higher required returns and expected distress costs
3. In the WACC approach, leverage affects value through the discount rate
4. In the APV approach, leverage affects value through separately valued tax benefits and bankruptcy costs
5. Minimizing WACC maximizes value only if operating cash flows are unaffected by leverage
6. The optimal debt ratio is the one that maximizes firm value, not necessarily the one with the most debt
7. Capital-structure analysis is a recapitalization question: how much value is created or destroyed by moving from current leverage to target leverage?

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